

Color Recognition Grasping Broadcast

Note: Due to unavoidable servo mechanical errors and the influence of servo center position, if the grabbing and placing positions are not ideal, you can try recalibrating the robotic arm or modify the grabbing and placing positions according to your needs.

1. Function Description

The voice module broadcasts the recognized color of the block, and then the robotic arm grabs the block of the corresponding color and places it in the set position.



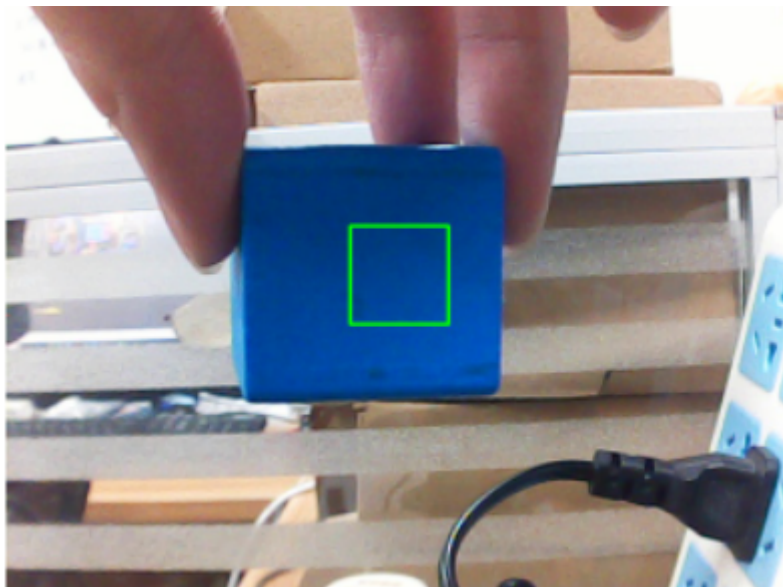
2. Startup and Operation

Open the terminal and enter the following command to start:

```
python3 ~/dofbot_voice/scripts/color_grab_broadcast.py
```

Operation Steps

After the program runs successfully, hold the color block and place the block within the green rectangle in the image, as shown in the figure below:



The program will recognize the block color. After the buzzer beeps once, you need to quickly place the block in the corresponding area (if it is a blue block like in the tutorial, place it in the blue area). The voice module will broadcast the recognized block color. Go to the corresponding color area to grab the block and place it in the middle gray-black area. Finally, the voice module will broadcast "Placement complete", and the terminal will print the recognized color.

```
yahboom@yahboom-virtual-machine:~$ python3 ~/dofbot_voice/scripts/color_grab_broadcast.py
Speech Serial Opened! Baudrate=115200
/home/yahboom/dofbot_voice/scripts/color_grab_broadcast.py:229: DeprecationWarning: setDaemon() is deprecated, set the daemon attribute instead
  thread1.setDaemon(True)
/home/yahboom/dofbot_voice/scripts/color_grab_broadcast.py:186: DeprecationWarning: setDaemon() is deprecated, set the daemon attribute instead
  closeTid.setDaemon(True)
red
blue
red
red
red
blue
green
yellow
red
yellow
yellow
```

3. Core Code Analysis

Source code path: `~/dofbot_voice/scripts/color_grab_broadcast.py`

```
# Call get_color to recognize the current block color
frame, color_name = get_color(frame)
# According to the recognized block color, pass the corresponding parameters to
the start_move_arm function, which will execute based on the recognized block
color
if len(color_name)==1:
    if color_name['name'] == 'yellow':
        start_move_arm(1)
    elif color_name['name'] == 'red':
        start_move_arm(2)
```

```

elif color_name['name'] == 'green':
    start_move_arm(3)
elif color_name['name'] == 'blue':
    start_move_arm(4)

# Number function definition
def number_action(index):
    if index == 1:
        # Grab yellow block
        arm_move(p_top, 1000)
        arm_move(p_Yellow, 1500)
    elif index == 2:
        # Grab red block
        arm_move(p_top, 1000)
        arm_move(p_Red, 1500)
    elif index == 3:
        # Grab green block
        arm_move(p_top, 1000)
        arm_move(p_Green, 1500)
    elif index == 4:
        # Grab blue block
        arm_move(p_top, 1000)
        arm_move(p_Blue, 1500)
    arm_clamp_block(1)
    arm_move(p_top, 1500)

def ctrl_arm_move(index):
    arm_clamp_block(0)
    if index == 1:
        print("Yellow")
        mySpeech.void_write(66)
        time.sleep(0.1)
        Arm.Arm_Buzzer_On(1)
        time.sleep(.5)
        number_action(index)
        put_down_block()
        mySpeech.void_write(65)
        time.sleep(0.1)
    elif index == 2:
        print("Red")
        mySpeech.void_write(69)
        time.sleep(0.1)
        Arm.Arm_Buzzer_On(1)
        time.sleep(.5)
        number_action(index)
        put_down_block()
        mySpeech.void_write(65)
        time.sleep(0.1)
    elif index == 3:
        print("Green")
        mySpeech.void_write(67)
        time.sleep(0.1)
        Arm.Arm_Buzzer_On(1)
        time.sleep(.5)
        number_action(index)
        put_down_block()
        mySpeech.void_write(65)
        time.sleep(0.1)

```

```
elif index == 4:  
    print("Blue")  
    mySpeech.void_write(67)  
    time.sleep(0.1)  
    Arm.Arm_Buzzer_On(1)  
    time.sleep(.5)  
    number_action(index)  
    put_down_block()  
    mySpeech.void_write(65)  
    time.sleep(0.1)
```